When Agreement Is Not Evidence

Measuring correlated failure in multi-agent visual evidence resolution

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The problem

A team of drones inspects a road for a blockage. Drone A reports a blockage. Drone B independently confirms it. The system closes the inspection task, because two sources agreed.

That inference is only valid if the two reports are independent. If both drones run the same detector, view the same scene through the same occluder, or operate under the same lighting, their errors are not independent. They can agree and be wrong together. Agreement then looks like evidence while carrying almost none.

This is not a hypothetical concern for this project. The existing prototype encodes the assumption directly, in code written without anyone noticing the problem:

```
# src/evidence_resolution/policy.py
has_independent_positive = any(
    item.source_id != claim.source_id for item in snapshot.positive_claims
)

if claim.assertion == "present" and has_independent_positive: return PolicyDecision( belief_stat
```

Different source identity is taken to mean statistical independence, and a task is closed on that basis. The reason code says `independent_confirmation`, but nothing establishes independence. This is what a competent engineer writes by default, which is precisely why it is worth measuring. The project's own accepted constraint AD-004 states that correlated or repeated evidence must not be treated as independent, but nothing in the system currently tests when that assumption breaks or what it costs when it does.

The gap

A 2026 survey of collaborative multi-agent perception states the problem and states that it is unmeasured. The source for both quotations below is *General Collaborative Intelligence: Architecting Cognition for Resilient Multi-Agent Ecosystems* (arXiv:2609.22967, 2026). On correlated failure, that survey states:

> "homogeneous agents with identical perception models and similar viewpoints are susceptible to correlated failures, where all agents miss the same occluded entity; collaboration does not help when the collective shares a common blind spot."
>
> — *General Collaborative Intelligence*, arXiv:2609.22967

And on the state of evaluation, the same survey describes the case where a fused collective model becomes worse than what each agent would achieve independently:

> "this phenomenon is documented in early work on Byzantine-resilient fusion" ... "but is rarely evaluated as a benchmark metric."
>
> — *General Collaborative Intelligence*, arXiv:2609.22967 (the elision is a citation marker)

A bounded primary-source search supports that assessment for this domain. RCP-Bench (CVPR 2025) benchmarks collaborative perception robustness under sensor corruptions such as noise and weather, not under correlated failure between agents. STAMP (2025) observes a weaker agent degrading collective performance, but reports it as a case observation rather than defining a measure. The existing cross-agent metric, Average Recall of Collaborative View, scores what collaboration adds, not what redundancy conceals.

The phenomenon has been quantified outside perception. Work on correlated agreement blindness in multi-agent triage reports a joint error rate inflated 3.53 times over what independence predicts, on network intrusion detection and clinical data using classical models. Parallel results exist in multi-agent language model research. Neither line addresses visual multi-agent perception, and neither releases a reusable benchmark.

What we propose to build

A benchmark and measurement harness for correlated failure in multi-agent visual evidence resolution, reusing the validated design of the existing prototype (see Feasibility below) rather than assuming its code carries over intact.

1. **A scenario generator with controlled dependence.** Scenarios are produced from a latent scene model so that the degree and the *source* of correlation between agents are set deliberately rather than observed accidentally. The planned correlation channels are shared occluder, shared detector model version, shared viewpoint geometry, and shared illumination.
2. **A dependence measure computed from claim streams alone.** The primary candidate is a dependence inflation factor: the observed joint error rate between two sources divided by the joint error rate independence would predict. It requires no cross-covariance and no access to simulator truth at resolution time.
3. **An agreement-reliability measure.** The probability that a claim is correct given that two or more sources agree, reported as a function of the dependence inflation factor. This is the measure the project actually needs, because agreement is the signal every fusion policy consumes.
4. **Task-outcome scoring.** Collaboration gain relative to the best single agent, plus false task closure, incorrect reopening, and verification cost. Simulator truth is used for scoring only and never enters resolution, per AD-001.

What the experiment would show

The study sweeps dependence from independent to fully correlated across the four correlation channels and reports where the independence assumption stops paying for itself. Concretely, it answers:

- At what dependence level does a second confirming report stop reducing the false-closure rate?
- Does the dependence inflation factor, computed without ground truth, predict that breakdown well enough to act on?
- Do the four correlation channels degrade decisions equivalently, or does shared model version behave differently from shared viewpoint?

The outcome is a characterization and a reusable instrument, with a stated operating boundary for the naive-independence baseline and its successors.

What we are not claiming

- Not a new fusion operator, and not an improvement on covariance intersection or any existing conservative fusion method.
- Not the discovery of correlated failure. The phenomenon is established in other domains; this work ports it to collaborative visual perception and supplies the missing measurement.
- Not a formal safety, calibration, or real-world operational claim. Results will establish decision-policy behavior in simulation only.

- Not a field-first claim. The novelty assessment is bounded search evidence. Its method, its two rounds, and the errors the adversarial pass caught are documented below and in the accompanying document `DEPENDENCE_NOVELTY_SURVEY.md`.

How this framing was tested before proposing it

This proposal is the second framing considered, not the first. Both were put through an adversarial novelty pass before reaching you. The full record is in the accompanying documents `DEPENDENCE_NOVELTY_SURVEY.md` and `DIRECTION_STATUS.md`.

Method. Primary-source retrieval, with every quotation verified verbatim against a locally retained copy of its source using a regex validation script. Claims directly asserted by a source are separated from inferences derived here. Findings were then handed to an independent adversarial reviewer instructed to attack rather than approve, and every objection it raised was re-verified against the source text before being accepted.

Round 1 — the first framing failed. The original candidate was online estimation of inter-source dependence, scored on task-lifecycle outcomes. Retrieval found the estimation half occupied: an IEEE TKDE paper from January 2026 performs HMM-based estimation of dynamic source dependence from categorical conflicting reports. A 2026 multi-robot maritime system was found to already combine visual evidence, unknown-correlation fusion, and tasking by expected uncertainty reduction, which removed the novelty of the proposed value-of-information gate.

The adversarial pass rejected the base survey. The literature survey was drafted first and is referred to here as the base survey. It was submitted to an independent adversarial reviewer agent, which was instructed to attack the survey's reasoning rather than approve it, and which returned a verdict of NEEDS REVISION with five blocking issues. Four were re-verified directly against the retained source text and corrected in the survey:

1. The base survey claimed that a counterexample system had no task lifecycle. The source text contradicts this. That system implements track management, an assign and reassign state machine, an explicit anti-chattering term, and evaluation on fragmentation and identity continuity. Its exclusion was rewritten to rest on the single ground that survives: it bounds correlation conservatively rather than estimating it.
2. The base survey described a source's method as "online" and labelled that a source claim. The source claims "dynamic," not "online," and extracts dependence via the Viterbi algorithm, which decodes a complete observation sequence. "Online" was an inference presented as an assertion.
3. The base survey dispositioned a 2026 survey as supportive motivation. That survey in fact proposes its own benchmark and describes independence-conditioned acceptance gating, which

makes it a threat to the claim rather than support for it. It was re-dispositioned accordingly.
4. The base survey made negative claims about a benchmark from its abstract page alone, without disclosing the partial read. The disclosure was added.

Two quotations also failed verbatim validation, caused by mangled mathematical notation and a citation bracket in the retrieved text. Both were restated in verified wording. The full record, including the corrections, is retained in the accompanying document `DEPENDENCE_NOVELTY_SURVEY.md`.

Round 2 — the present framing survived. The same process was applied to this proposal's framing. The gap is asserted by a retrieved source rather than inferred. The nearest threat, a 2026 paper on correlated agreement blindness reporting joint error inflated 3.53 times over independence, was checked and operates on network intrusion detection and clinical tabular data with classical models, not visual perception, and defines indicators rather than releasing a benchmark.

The stated limitations of that assessment are recorded with it: a two-year source recency gate that functions as a search budget rather than a novelty boundary, three sources read only in part, and general web indexes rather than a systematic database protocol. It is bounded search evidence, not proof of absence.

Honest limitation

The contribution is a domain transfer plus an instrument, not a new phenomenon. Correlated failure among agents is understood elsewhere. The defensible claims are that it is unmeasured in collaborative visual perception, that the measurement is constructible, and that the resulting numbers change how an evidence-gating policy should be designed. If a reviewer finds a published benchmark that measures failure correlation against task outcomes in multi-agent perception, this framing should be withdrawn or narrowed.

Feasibility

A working prototype exists: typed claim and trace contracts, a replaceable policy interface, four oracle scenarios, deterministic replay verified by matching SHA-256 trace hashes, and a command-line harness. It was built in a single session. It is treated as disposable, and carries no frozen status.

What it establishes is that the design is sound, and three design decisions carry forward independently of the code that currently implements them:

1. **A single policy seam.** Comparison policies plug in behind one interface, which is what makes a baseline sweep cheap to run.

2. **A naive-independence baseline.** A policy that treats a differing source identifier as statistical independence is exactly the assumption this study measures, so one will exist in the comparison set. The prototype shows that rule is what a reasonable engineer writes by default, which is the point.
3. **Separation of simulator truth from the agent-visible claim stream**, on which the whole evaluation depends (AD-001).

The implementation itself is expected to be rebuilt rather than extended, for concrete reasons. The current claim record carries a single observation timestamp and no arrival time, so it cannot express late or out-of-order delivery, which this study must control for in order to avoid confounding ordering effects with dependence effects. The four existing scenarios exercise support, contradiction, and staleness; none exercises correlation between agents, so a new corpus is required regardless.

This is a deliberate position rather than a concession. The project's risk was never implementation volume; it is whether the measurement is well defined. A prototype this small is cheaper to rebuild against a clear specification than to bend toward a purpose it was not designed for, and rebuilding removes any pressure to preserve decisions that were convenient rather than correct.

The work requires no new simulator dependency, no flight hardware, and no perception training. The PX4 and Gazebo spike remains optional and off the critical path.

Requested decision

Approval to proceed to a design phase for this framing: freeze the dependence measure, the correlation channels, the scenario matrix, and the acceptance thresholds. Implementation remains a separate approval gate.